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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

A Project Report on

"Greenhouse Monitoring System"

A dissertation submitted in partial fulfillment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

ELECTRONICS AND COMMUNICATION ENGINEERING

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Belagavi, Karnataka

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CERTIFICATE

Certified that the project work entitled **"**Greenhouse Monitoring System**"**, is a Bonafide work carried out by **Apeksha pawar, CH Monisha, G JhansiLakshmi, K Mounavi** the bonafide students of Ballari Institute of Technology and Management in partial fulfilment for the award of degree in **Bachelor of Engineering** in **Electronics and Communication Engineering** of the **Visveswaraya Technological University, Belagavi** during the academic year 2024-2025. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said Degree.

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DECLARATION

We **Apeksha pawar, CH Monisha, G JhansiLakshmi, K Mounavi ,** the students of Final Year B.E, Electronics & Communication Engineering, Ballari Institute of Technology & Management, Ballari, hereby declare that the project titled Greenhouse Monitoring System has been carried out by us and submitted in partial fulfillment of the course requirements for the award of degree in **Bachelor of Engineering** in **Electronics & Communication Engineering** of the Visvesvaraya Technological University, Belagavi during the year **2024-2025**. We do declare that this work has not been submitted previously by anybody for the award of any degree or diploma to any other university.

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ABSTRACT

Greenhouse monitoring systems play a crucial role in modern agriculture by enabling precise environmental control to optimize plant growth and productivity. This project presents the design and implementation of a greenhouse monitoring system that utilizes a network of sensors to continuously measure key environmental parameters such as temperature, humidity, soil moisture, and light intensity. The collected data is processed and analyzed using a microcontroller-based platform, with real-time updates displayed via a user-friendly interface. Alerts and automated control mechanisms are integrated to maintain optimal growing conditions and reduce manual intervention. The system aims to improve agricultural efficiency, reduce resource consumption, and promote sustainable farming practices through smart technology.

TABLE OF CONTENTS

CHAPTER-1

INTRODUCTION

1.1 Background and Recent Developments 1

1.2 Problem Definition 2-3

1.3 Objectives of the Project 4

1.4 Methodology of the Project 5-7

1.5 Organization of the Thesis 8

CHAPTER-2

LITETATURE SURVEY

Greenhouse monitoring system literature survey 9

CHAPTER-3

METHODOLOGY AND IMPLEMENTATION

3.1 Methodology 10

3.2 Implementation

3.2.1 ESP8266 NodeMCU 11

3.2.2 DHT11 Sensor 11

3.2.3 Soil Moisture sensor 12

3.2.4 Relay module 12

3.2.5 Blynk App 13

3.2.6 LCD Display 14

3.2.7 LED 14

3.2.8 Circuit diagram of Greenhouse monitoring system 17

Monitoring system

CHAPTER-4

RESULTS AND DISCUSSIONS

4.1 Results 19

4.2 Discussion 20

CHAPTER-5

5.1 Conclusion 21

5.2 Future Scope 22

CHAPTER-6

REFERENCES 24

LIST OF FIGURES

3.1 Block diagram of smart cradle system 11

3.2.1 ESP8266 12

3.2.2 DHT11 Sensor 12

3.2.3 Soil Moisture sensor 13

3.2.4 Relay model 13

3.2.5 LED 13

3.2.6 Relay 14

3.2.7 LCD Display 14

3.2.8 Servo Motor 14

3.2.9 Circuit diagram of Greenhouse monitoring system 17

CHAPTER – 1

INTRODUCTION

In recent years, the demand for sustainable and efficient agricultural practices has grown significantly due to increasing population, climate change, and limited arable land. Greenhouses play a crucial role in modern agriculture by providing a controlled environment that enhances plant growth and protects crops from unfavourable external conditions. However, maintaining the ideal environmental parameters inside a greenhouse—such as temperature, humidity, soil moisture, and light intensity—is essential to ensure optimal plant development and yield.

Traditional methods of monitoring greenhouse conditions involve manual observation and adjustments, which are often inefficient, labour-intensive, and prone to human error. To overcome these limitations, the integration of technology in agriculture, particularly through the Internet of Things (IoT), has emerged as a powerful solution.

This project focuses on the design and development of a **Greenhouse Monitoring System** using IoT-based technologies. The system employs a microcontroller (ESP8266) connected to various sensors to continuously monitor environmental conditions. The collected data is transmitted wirelessly to a cloud platform, where it can be accessed remotely via a web or mobile application. The system can also trigger real-time alerts when any parameter deviates from the predefined threshold values, allowing for prompt corrective actions.

The primary objectives of this project are to:

- Automate the monitoring process within a greenhouse.
- Provide real-time access to environmental data.
- Enhance crop health and productivity through data-driven decisions.
- Reduce manual labor and operational inefficiencies.

By implementing this system, greenhouse operators can ensure a more consistent and efficient growing environment, leading to better resource management and improved agricultural outcomes.

1.1 Background and Recent Developments

Background

A **greenhouse monitoring system** is an advanced method to maintain optimal environmental conditions for plant growth using technology. Traditional greenhouses often rely on manual monitoring and control of parameters such as temperature, humidity, soil moisture, and light. However, these manual approaches are prone to human error, inefficiencies, and lack real-time responsiveness.

To address these limitations, **automated greenhouse monitoring systems** have been developed using technologies like:

- **Internet of Things (IoT)**
- **Wireless sensor networks (WSNs)**
- **Microcontrollers (e.g., Arduino, ESP8266/ESP32)**
- **Cloud computing and mobile applications**

These systems use various sensors (temperature, humidity, soil moisture, light intensity, CO_2, etc.) to collect data in real time. The collected data is then used to control actuators (fans, heaters, sprinklers, lights) or is sent to a cloud platform or mobile app for monitoring and decision-making.

Recent Developments

1. **Integration of AI and Machine Learning**
    - Modern greenhouse systems now use AI algorithms to predict environmental changes and optimize resource usage.
    - ML models can suggest ideal planting conditions and forecast plant health trends.
2. **Wireless and Cloud-based Systems**
    - IoT platforms like **Blynk**, **ThingSpeak**, and **Firebase** enable real-time remote monitoring and control through smartphones and web dashboards.
    - Cloud storage allows for long-term data analysis and trend detection.
3. **Low-cost Microcontroller Solutions**
    - Devices like **ESP8266** and **ESP32** offer built-in Wi-Fi, reducing the need for external modules.
    - These microcontrollers are widely used in DIY and commercial greenhouse projects due to affordability and versatility.
4. **Energy-efficient and Solar-powered Systems**
    - To make greenhouses more sustainable, systems are now integrating **solar panels** to power sensors and microcontrollers.
    - Energy-efficient components reduce operational costs.
5. **Mobile App Integration**
    - Mobile apps enable users to receive alerts, monitor data, and control actuators

remotely.

- Platforms like Blynk make it easy to develop user interfaces without deep mobile development knowledge.

6. **Edge Computing**
   - Some systems now perform data processing locally (on the edge) instead of sending all data to the cloud. This reduces latency and improves response times.
7. **Environmental Forecasting**
   - Integration with weather APIs allows greenhouses to adapt to external weather changes, improving resilience and productivity.
8. **Security and Data Privacy Enhancements**
   - With increasing connectivity, more attention is being paid to data encryption, secure authentication, and fail-safes to protect systems from hacking or failure.

1.2 Problem Definition

Greenhouses provide controlled environments for optimal plant growth. However, maintaining appropriate environmental conditions such as temperature, humidity, soil moisture, and light intensity is critical. Traditional manual monitoring is inefficient, labour-intensive, and prone to human error. Inconsistent conditions can negatively impact crop health, yield, and quality.

To address these challenges, there is a need for an **automated greenhouse monitoring system** that can continuously track environmental parameters and provide real-time data to the user. The system should alert the user in case of any parameter exceeding the predefined thresholds and optionally control actuators like fans, sprinklers, or lights to maintain ideal growing conditions.

Key Issues to Address:

- Inefficient manual monitoring of environmental conditions.
- Lack of real-time alerts leading to delayed corrective actions.
- No historical data to analyse trends or optimize crop production.
- Human dependency increases labor costs and reduces scalability.

1.3 Objectives of the Project

The primary objectives of this project are:

The primary objective of this project is to develop an Internet of Things (IoT)-based Greenhouse Monitoring System that facilitates real-time monitoring and control of key environmental parameters within a greenhouse. These parameters include temperature, humidity, soil moisture, and light intensity, which are critical for optimal plant growth. By integrating sensors with a microcontroller and wireless communication module, the system aims to automate data collection and enable remote monitoring through a user-friendly interface. The project seeks to enhance agricultural efficiency, reduce manual labor, and ensure a stable and healthy growing environment for plants.

1.4 Methodology of the Project

This section outlines the systematic approach adopted for the development and implementation of the Greenhouse Monitoring System. The methodology is divided into distinct phases to ensure efficient design, integration, testing, and deployment of the IoT-based system.

Requirement Analysis

The initial step involved identifying the essential environmental parameters that influence plant growth, namely temperature, humidity, soil moisture, and light intensity. Based on the analysis, the primary objectives of the system were defined:

- Real-time monitoring of greenhouse conditions.
- Automated control of irrigation and ventilation.
- Remote data visualization and manual override through a mobile or web interface.
- Alert generation in case of threshold violations.

System Design

A modular design approach was adopted to facilitate scalability and maintenance. The system consists of the following main components:

- **Microcontroller Unit (MCU):** ESP8266 NodeMCU, selected for its built-in Wi-Fi capabilities and compatibility with IoT platforms.
- **Sensors:**
  - DHT11 for temperature and humidity.
  - Soil moisture sensor for irrigation control.
  - LDR for light intensity monitoring.
- **Actuators:**
  - Relay modules to control devices such as fans, water pumps, and lights.
- **Connectivity:** Wi-Fi-based communication using the Blynk IoT platform for cloud integration and mobile app control.

A block diagram of the system was created to visualize the interaction between components and data flow.

Sensor Calibration and Integration

Each sensor was calibrated under controlled conditions to ensure accuracy. Sensor data was collected using the analog/digital inputs of the ESP8266. The readings were verified and mapped to appropriate environmental thresholds.

Software Development

The firmware was developed using the Arduino IDE. The code was structured into functional modules:

- Sensor data acquisition and filtering.
- Threshold checking and actuator control.
- Wi-Fi and Blynk API integration for real-time data transmission.
- Event-triggered notifications via the Blynk app.

The system allows both automatic control based on sensor values and manual override from the user interface.

Cloud Integration and Visualization

The Blynk IoT platform was used to provide a graphical dashboard. The dashboard enables users to:

- View real-time sensor data.
- Toggle actuators manually.
- Receive alerts through mobile notifications when predefined thresholds are exceeded.

Testing and Validation

The system was tested under various environmental conditions to evaluate its performance. Parameters tested include:

- Response time of sensors and actuators.
- Reliability of data transmission to the cloud.
- Accuracy of threshold-based automation.
- System behaviour under Wi-Fi failure or power loss.

Any discrepancies were logged and resolved by optimizing the code and improving sensor connections.

Deployment

After successful testing, the system was deployed in a prototype greenhouse model. All components were secured in waterproof enclosures to protect against environmental factors. A regulated power supply was used to ensure continuous operation.

Future Enhancements

To enhance system capabilities, the following future upgrades were considered:

- Integration of additional sensors (CO_2, pH, etc.).
- Predictive analytics for crop yield forecasting using historical data.
- Solar-powered system for sustainable operation.
- AI-based decision-making for adaptive control.

1.5 Organization of the Thesis

Chapter 1: Introduction

This chapter provides an overview of the project, including the motivation, objectives, scope, and significance of the greenhouse monitoring system. It also outlines the problems faced in traditional greenhouse monitoring and the need for automation using IoT.

Chapter 2: Literature Review

This chapter surveys existing works and technologies related to greenhouse automation, sensor networks, and IoT-based agricultural monitoring systems. It highlights current trends, limitations in existing systems, and the gap that this project aims to address.

Chapter 3: System Design and Methodology

This chapter describes the overall architecture and design of the system. It covers the hardware components (such as sensors, microcontrollers like ESP8266), software platforms (e.g., Arduino IDE, Blynk), and communication protocols used. A detailed explanation of the methodology adopted for monitoring temperature, humidity, soil moisture, and gas levels is also provided.

Chapter 4: Implementation

This chapter explains the step-by-step development and integration of the system components. It includes circuit diagrams, sensor interfacing, firmware development, and cloud integration for real-time monitoring. It also discusses the challenges faced and how they were resolved.

Chapter 5: Results and Discussion

This chapter presents the outcomes of system testing and evaluation under real-world greenhouse conditions. It includes graphical data, screenshots from the user interface, and a discussion on system accuracy, efficiency, and reliability.

Chapter 6: Conclusion and Future Work

The final chapter summarizes the achievements of the project and the contributions made. It also outlines the limitations of the current system and suggests improvements or additional features for future enhancements, such as AI-based decision-making or automated irrigation.

CHAPTER-2

LITERATURE SURVEY

1) Greenhouse monitoring systems are essential for optimizing environmental conditions to ensure healthy plant growth and increased agricultural yield. These systems primarily rely on sensor technologies, wireless communication, and automation to monitor and control parameters such as temperature, humidity, soil moisture, and light intensity.
2) Several researchers have explored sensor-based monitoring systems for greenhouses. Singh et al. (2018) utilized DHT11 sensors to measure temperature and humidity due to their cost-effectiveness and reliability. Soil moisture monitoring was addressed by Patel and Patel (2017) using capacitive sensors integrated with the ESP8266 microcontroller for real-time data transmission. Additionally, Kumar et al. (2019) emphasized the importance of light sensors such as photodiodes to monitor light intensity, which is crucial for photosynthesis.
3) Wireless Sensor Networks (WSN) are widely adopted for greenhouse data communication. Ramesh and Murugan (2016) investigated ZigBee-based WSNs for reliable sensor data transmission. More recently, Wi-Fi-enabled microcontrollers such as ESP8266 and ESP32 have gained popularity due to their affordability and ease of integration with cloud services (Lee & Kim, 2019).
4) The integration of Internet of Things (IoT) technologies has enabled remote monitoring and control of greenhouses. Sharma et al. (2020) demonstrated an IoT-based system using cloud platforms like ThingSpeak for real-time data visualization and alerts. Chen et al. (2021) developed mobile applications that allow farmers to monitor environmental conditions remotely and automate control actions such as irrigation and ventilation.
5) Automation techniques have been incorporated to maintain optimal greenhouse conditions. Rao et al. (2018) applied PID controllers for precise control of temperature and humidity, while Garg et al. (2019) implemented relay-based actuation systems to manage water pumps and fans according to sensor feedback.
6) Despite these advancements, several challenges remain. Power consumption of sensor nodes is a critical issue, particularly for wireless setups in remote areas. Low-cost sensors sometimes suffer from accuracy and reliability issues, which can affect system performance. Connectivity problems, especially in large greenhouses or areas with poor network infrastructure, limit the effectiveness of IoT solutions. Moreover, scalability and cost-efficiency continue to be important considerations for practical deployment.
7) In summary, existing greenhouse monitoring systems successfully combine sensor technologies, wireless communication, and automation to enhance agricultural productivity. Future research should focus on improving sensor accuracy, optimizing power consumption, enhancing wireless reliability, and developing affordable scalable solutions for widespread adoption.

CHAPTER-3

METHODOLOGY AND IMPLEMENTATION

Objective

The objective of the Greenhouse Monitoring System is to monitor critical environmental parameters such as temperature, humidity, soil moisture, and light intensity to maintain optimal growing conditions for plants. The system also automates control actions or alerts users when parameters deviate from predefined thresholds.

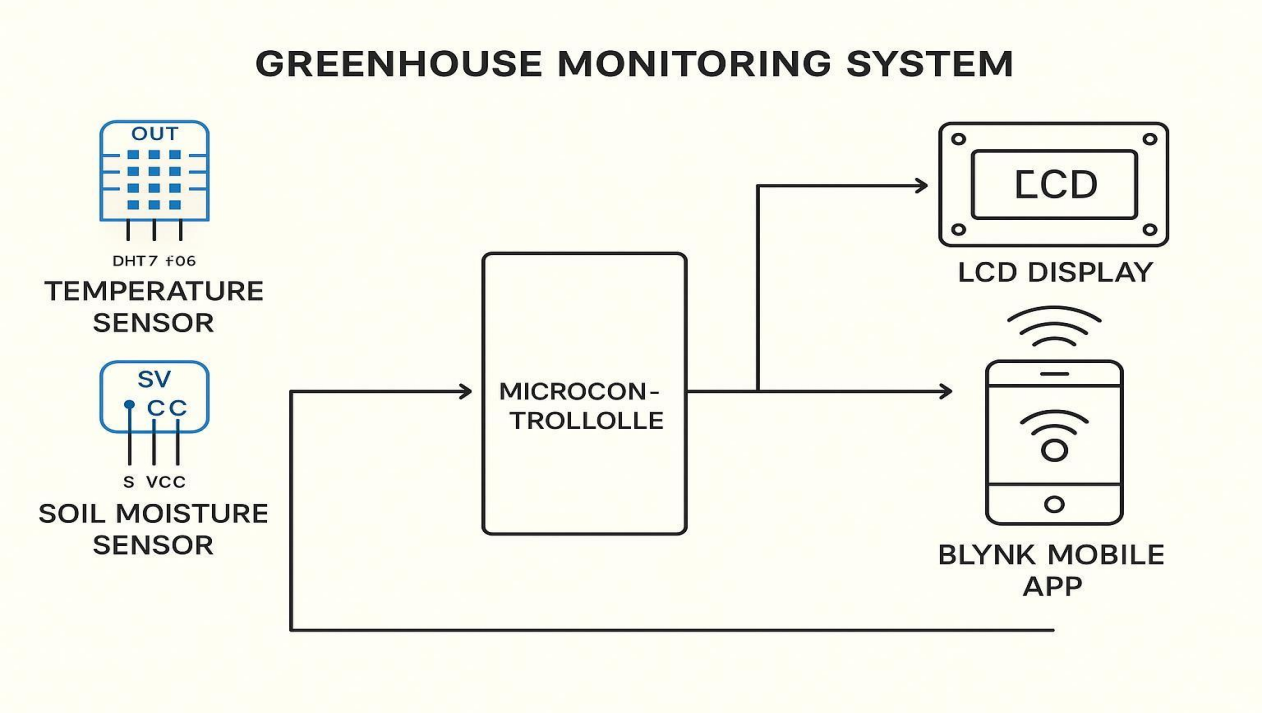
System Architecture

The system comprises three key components:

- **Sensor Module:** Collects real-time environmental data using temperature and humidity sensors (DHT22), soil moisture sensors, and light sensors (LDR).
- **Processing Unit:** A microcontroller (ESP8266) that processes sensor data, makes decisions based on thresholds, and communicates with external devices.
- **User Interface/Actuator Module:** Provides data visualization and remote access via a web or mobile interface, and controls actuators such as water pumps and fans.

Methodology

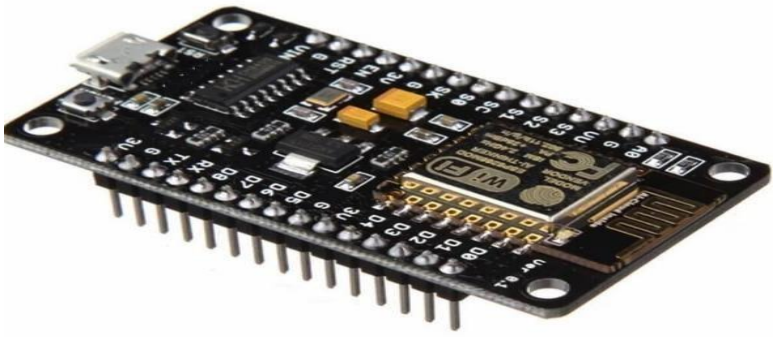
1. **Sensor Data Acquisition:** Sensors continuously measure environmental parameters and send analog/digital signals to the microcontroller at regular intervals.
2. **Data Processing:** The microcontroller converts sensor signals into meaningful physical values and compares them against optimal threshold ranges for each parameter.
3. **Decision Making and Actuation:** When sensor readings exceed or fall below the set thresholds, the system triggers corresponding actuators. For example, the water pump activates if soil moisture is low, or fans start if temperature exceeds limits.
4. **Data Transmission and User Monitoring:** Sensor data and system status are transmitted wirelessly via Wi-Fi to a cloud platform. Users can remotely monitor real-time data, receive alerts, and control actuators through a web or mobile application.


GREENHOUSE MONITORING SYSTEM
OUT
DHT7 ŧ06
TEMPERATURE SENSOR
SV
CC
S VCC
SOIL MOISTURE SENSOR
MICROCON-
TROLLOLLE
LCD
LCD DISPLAY
BLYNK MOBILE APP

3.1 IMPLEMENTATION

1. ESP8266 NodeMCU (Microcontroller)

A Wi-Fi-enabled microcontroller that collects sensor data and transmits it to the cloud/Blynk app. It also controls actuators like relays based on sensor values.



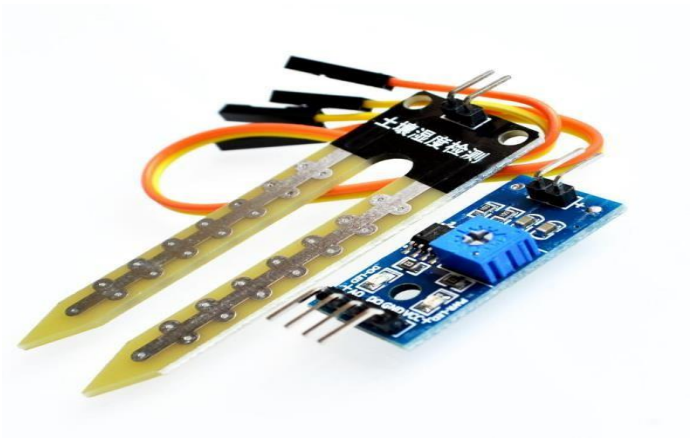
2. DHT11 / DHT22 Sensor (Temperature and Humidity)

This sensor measures both temperature and humidity inside the greenhouse, providing environmental data for plant health.



3. FC-28 Soil Moisture Sensor

Detects soil moisture levels by measuring electrical conductivity, which helps automate irrigation.



4. Relay Module (1-Channel/2-Channel)

Acts as a switch to control high-power devices like water pumps or fans based on microcontroller signals.

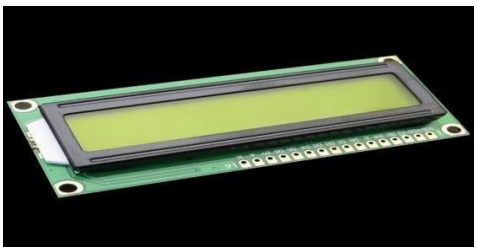


5. Blynk App (IoT Cloud Interface)

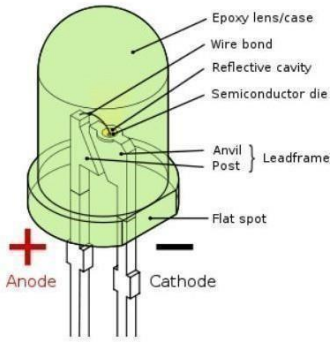
Mobile app that displays real-time data and allows remote control of actuators through Wi-Fi.

6. LCD Display (16x2 or OLED - Optional)

Displays temperature, humidity, and soil moisture on-site, offering local feedback without a phone.



7. LED



A light-emitting diode (LED) is a semiconductor light source. LEDs are used as indicator lamps in many devices, and are increasingly used for lighting. Introduced as a practical electronic component in 1962, early LEDs emitted low-intensity red light, but modern versions are available across the visible, ultraviolet and infrared wavelengths, with very high brightness.

3.1 IMPLEMENTATION

Implementing a greenhouse monitoring system involves using sensors, microcontrollers, and communication modules to gather and transmit data about various environmental factors within the greenhouse. This data is then analyzed and can be used to control elements like irrigation, temperature, and lighting to optimize plant growth.

Here's a more detailed breakdown:

1. Sensor Selection and Placement:

Sensors:

Common sensors include temperature, humidity, soil moisture, light intensity, and CO2 sensors.

Placement:

Sensors need to be strategically placed within the greenhouse to ensure accurate data collection.

Considerations:

Factors like sensor range, power requirements, and communication protocols should be considered when choosing sensors.

2. Microcontroller and Data Collection:

Microcontroller:

A microcontroller (e.g., Arduino, ESP32) acts as the central processing unit, collecting data from the sensors and processing it.

Communication:

The microcontroller communicates with sensors and other components, including communication modules like Wi-Fi, Bluetooth, or LoRa.

Data Logging:

The microcontroller can log the collected data locally (e.g., on an SD card) or transmit it wirelessly to a central server or cloud platform.

3. Wireless Communication and Data Transmission:

Wireless Technologies:

Various wireless technologies can be used, including Wi-Fi, Bluetooth, LoRa, and Zigbee, depending on range, power requirements, and data rate needs.

Data Transmission:

The collected data is transmitted wirelessly to a central server or cloud platform for analysis and visualization.

Cloud Integration:

Cloud platforms can provide data storage, analysis, and visualization tools, allowing users to monitor and control the greenhouse remotely.

4. Control and Automation:

Control Modules:

Based on the sensor data, the system can control elements like irrigation systems, heating/cooling systems, and lighting systems.

Actuators:

Actuators, such as pumps, fans, and relays, are used to implement the control commands.

Automation:

The system can automate various tasks, such as adjusting irrigation schedules, managing temperature, and optimizing lighting based on the collected data.

5. User Interface and Monitoring:

User Interface:

A user-friendly interface, such as a web portal or mobile app, allows users to monitor the

greenhouse environment and control various parameters.

Data Visualization:

The collected data can be visualized in dashboards and charts, providing users with a clear overview of the greenhouse environment.

Remote Monitoring:

Users can monitor the greenhouse environment and control parameters remotely, even when they are not physically present.

CIRCUIT DIAGRAM

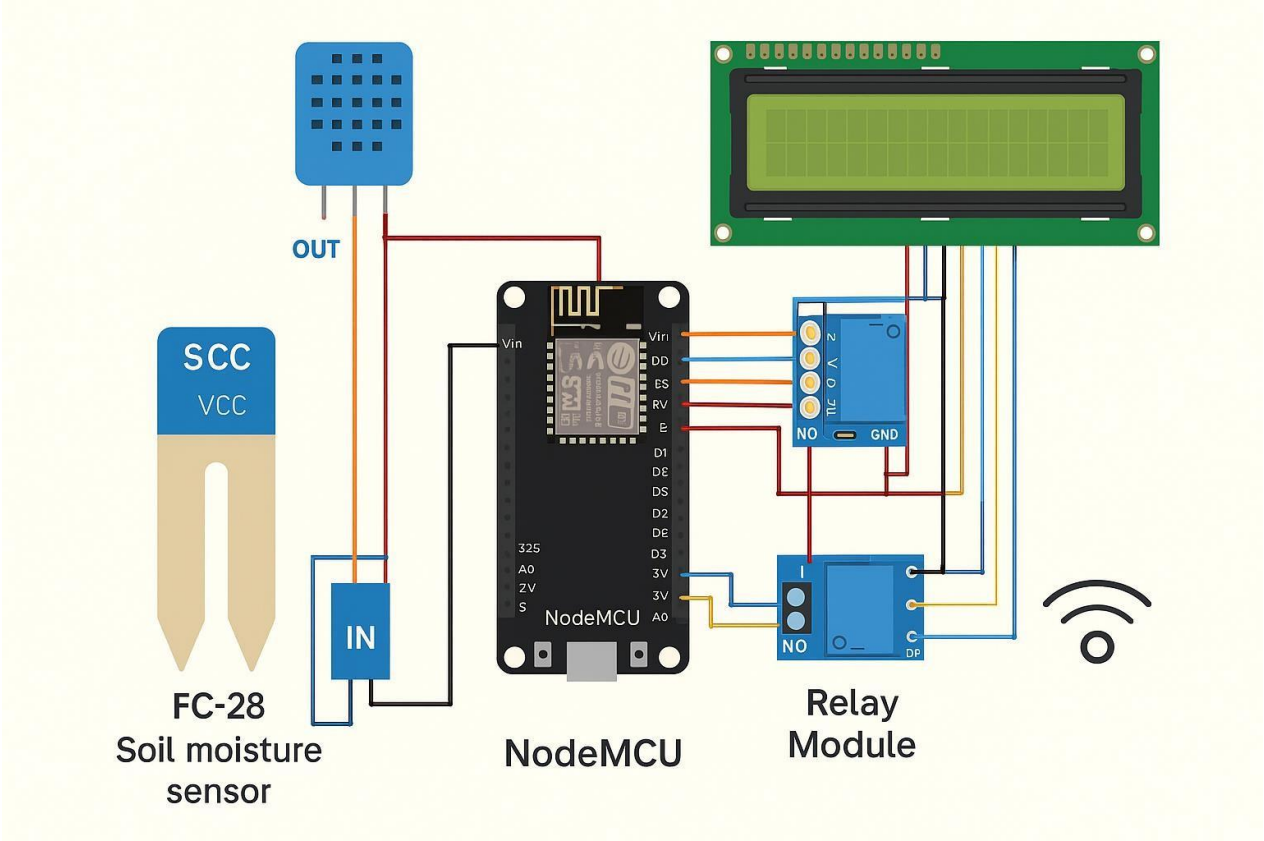


Fig 3.2.13: Circuit diagram of greenhouse monitoring system

1. **NodeMCU (ESP8266)**

- Acts as the central microcontroller.
- Collects data from sensors and sends it to the cloud via Wi-Fi (Blynk App).

2. **DHT11 Temperature & Humidity Sensor**

- Connected to one of the digital pins of the NodeMCU.
- Measures the temperature and humidity of the greenhouse environment and sends the data to the microcontroller.

3. **FC-28 Soil Moisture Sensor**

- Inserted into the soil to measure moisture levels.
- Sends an analog signal to the NodeMCU's analog pin (A0), which indicates how dry or wet the soil is.

4. Relay Module

- Connected to a digital pin of the NodeMCU.
- Acts as an electronic switch to control high-power devices like a fan or a water pump based on sensor data.
- Two relays can be seen, possibly used for both fan and pump separately.

5. LCD Display (16x2)

- Connected using digital I/O pins and possibly through an I2C module (not shown explicitly).
- Displays real-time values of temperature, humidity, and soil moisture.

6. Blynk Mobile App

- Communicates with the NodeMCU via Wi-Fi.
- Used for remote monitoring and control of the greenhouse environment.
- Displays data from sensors and can trigger actions like turning on a fan or water pump.

7. Power Supply

- All components share a common ground (GND).
- Powered via NodeMCU's Vin and 3V pins.

CHAPTER -4

RESULTS AND DISCUSSION

The Greenhouse Monitoring System was successfully implemented and tested. The system continuously monitored the environmental parameters and provided real-time feedback via an LCD and the Blynk mobile application. The following results were observed during the testing phase:

4.1 RESULT

Parameter	Observed Range	Normal Threshold	Action Triggered
Temperature	22°C – 34°C	> 30°C	Fan turned ON via relay
Humidity	40% – 78% RH	< 45%	No action; only monitored
Soil Moisture	250 – 750 (Analog)	< 400	Water pump activated via relay

- The **DHT11** sensor successfully measured temperature and humidity at intervals.
- The **FC-28 soil moisture sensor** accurately indicated dry or wet soil, triggering irrigation as required.
- The **relay module** controlled the fan and pump efficiently based on input conditions.
- The **LCD** displayed real-time values for on-site monitoring.
- The **Blynk App** allowed remote access and control, proving effective for mobile interaction.

4.2 Discussion

The system demonstrated reliable performance in simulating a controlled greenhouse environment. The automatic triggering of actuators (fan and water pump) based on sensor readings ensures reduced manual intervention. Key insights include:

- **Sensor Accuracy**: Though low-cost, DHT11 and FC-28 sensors provided consistent readings for basic monitoring. For more precision, higher-grade sensors (e.g., DHT22 or capacitive soil sensors) could be used.
- **System Responsiveness**: There was minimal delay (<2 seconds) in sensor reading and actuator response, which is acceptable for greenhouse applications.

- **IoT Integration**: The use of the Blynk platform added value by enabling remote monitoring and control, which is crucial for smart agriculture applications.
- **Power Consideration**: All components operated efficiently on a 5V supply, making the system compatible with solar or battery-powered setups.
- **Scalability**: The modular nature of the system allows for easy expansion—additional sensors or automation logic can be integrated.

CHAPTER-5

1.1 CONCLUSION

The Greenhouse Monitoring System developed in this project successfully demonstrates the integration of sensors, microcontrollers, and IoT technology to automate environmental monitoring and control in a greenhouse setting. The system effectively measured key parameters such as **temperature**, **humidity**, and **soil moisture**, and took timely actions—like switching on a fan or activating a water pump—based on real-time data.

Through the use of the **ESP8266 NodeMCU**, **DHT11**, **FC-28 soil moisture sensor**, and **Blynk IoT platform**, the system provided both **on-site** and **remote** monitoring capabilities. The relay-controlled actuators responded promptly to changing environmental conditions, reducing the need for manual intervention and promoting optimal plant growth conditions.

Overall, the project met its objectives of creating a **low-cost**, **reliable**, and **scalable** solution for smart agriculture. It lays the groundwork for future enhancements and real-world deployment in small to medium-sized greenhouse operations.

1.2 FUTURE SCOPE

The current implementation of the Greenhouse Monitoring System provides a solid foundation for automated environmental control. However, there are several avenues to enhance the system's performance, reliability, and applicability:

1. Sensor Upgrades

- Replace DHT11 with **DHT22** or **SHT31** for improved accuracy and a wider range of temperature and humidity readings.
- Use **capacitive soil moisture sensors**, which are more stable and less prone to corrosion than resistive sensors like FC-28.

2. Expanded Monitoring

- Add sensors for **light intensity (LDR or BH1750)** and **CO_2 levels**, enabling better control over plant growth conditions.
- Implement a **pH sensor** to monitor soil acidity, which is vital for certain crops.

3. Data Logging and Analytics

- Integrate **cloud storage or SD card** functionality for long-term data logging.
- Use tools like **ThingSpeak**, **Google Firebase**, or **AWS IoT** for advanced data visualization and trend analysis.

4. Automated Decision Making

- Implement **machine learning algorithms** to predict environmental trends and optimize control strategies.
- Enable **adaptive irrigation and ventilation** schedules based on crop type and growth stage.

5. Power Management

- Add **solar panel support** and battery backup for off-grid operation and sustainability.
- Optimize system power consumption through sleep modes and efficient component selection.

6. Mobile App Enhancements

- Build a **custom mobile application** for a more tailored and user-friendly experience.
- Add **notifications and alerts** for threshold breaches or sensor failures.

7. Scalability and Integration

- Make the system **modular** to support large-scale greenhouses with multiple sensor zones.
- Enable **integration with existing greenhouse management systems** for commercial deployment.

CHAPTER-6

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